

2nd Feb 2026 / CS6370

REC
SIGIR

⇒ IR Evaluation → Efficiency
→ Effectiveness

- Precision, Recall → shortcomings in IR Evaluation

- Efficiency → Time, Space, Cost

Index
a collecⁿ

Retrieval
time

human effort?
how to organize them?

Server Client
(Throughput, Parallelism) ↔ (Latency)

tradeoff

- how to handle curse of dimensionality in vector spaces in IR?

- document^{space} is sparse, it is in small clusters, why? effective space reduces drastically.

- there is no document which talks about everything, every document is specific

- effectiveness → precision, recall

↳ harder,

getting relevant docs.

(finds actual positives, or relevant docs.)

for evaluation is itself a challenge

Pooling → large collecⁿ of topics & relevant human judgement.

→ accuracy - contingency table

Ret	Y	N
ReE	TP	FN
Y		
N	FP	TN

there is a cost associated with this

- α -error $\rightarrow$ FP $\rightarrow$ retrieves sth. which is not relevant
 $\hookrightarrow \alpha \uparrow, FP \uparrow, Precision \downarrow$

β -error $\rightarrow$ fails to retrieve correct doc.
 (more critical)
 $\hookrightarrow FN \uparrow, \beta \uparrow, FN \uparrow, Recall \downarrow$

- PR is dependent on the application. What is actually sought?

relevant
want
query

$$Acc. = \frac{TP + TN}{TP + TN + FP + FN}$$

② for a rank $\rightarrow$ never used in IR sys. due to high class imbalance.

specially TN that does not make sense, most of docs are irrelevant.

- $S_1 \rightarrow R N N R R N N N N$
 $S_2 \rightarrow N N N N N N N$ } there are only 3 relevant docs.

$I \rightarrow R R R R R R R$

which sys. has higher acc.

$$S_1 = \frac{3 + 0}{3 + 4 + 0 + 0} = \frac{3}{7} \quad S_2 = 0$$

- Precision tends to Δ continuously. Recall is monotonically ever-increasing func.
 $\hookrightarrow$ It's general tendency is to reduce.
 - search engine produces a ranking, not a set